Highlights

Global Mg/Si and Al/Si maps of the Moon from Chandrayaan-2 CLASS with time-resolved solar excitation from GOES XRS

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- CLASS X-ray spectra fitted on raw counts with a Poisson likelihood
- GOES-temperature CHIANTI solar spectra used for per-exposure excitation
- Solar-driven instrumental Al K-alpha found in sunlit CLASS spectra
- Global Al/Si map validated against GRS ($r = 0.72$) and landing sites
- Ca, Ti and Fe need a measured solar spectrum; Mg/Si and Al/Si do not

Global Mg/Si and Al/Si maps of the Moon from Chandrayaan-2 CLASS with time-resolved solar excitation from GOES XRS

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Abstract

We present a processing chain for Chandrayaan-2 CLASS X-ray fluorescence spectra that is reproducible from public data. Each 8 s spectrum is fitted on raw counts with a Poisson likelihood, using Sherman primary fluorescence with matrix attenuation and incidence geometry, a scattered solar spectrum, a binned Gaussian detector response, the published filter transmission, and a particle-background template. The incident solar spectrum is an isothermal CHIANTI spectrum at the GOES XRS temperature of each exposure. Solar lines carry 55–80% of the flux that excites Mg, Al and Si, so element/Si ratios change by tens of per cent for realistic temperature errors. We also identify an instrumental Al Kα component that appears only when the Sun illuminates the spacecraft from behind; if it is mixed into the background model, it corrupts Al/Si. Synthetic tests recover Mg/Si and Al/Si without bias, while Ca/Si, Ti/Si and Fe/Si are not recoverable without a measured solar spectrum. From 1,515,239 exposures in 24 monthly archives, 578,530 spectra pass pre-declared quality criteria and give global 2° maps with quality layers and no infill. Al/Si resolves individual maria and correlates with Lunar Prospector GRS on 5° pixels ($r = 0.72$, about 80% of the noise-limited ceiling) and with nine landing-site soils ($r = 0.81$). Mg/Si correlates with GRS ($r = 0.61$); its landing-site test is inconclusive, as it is for GRS itself.

Keywords: Moon, X-ray fluorescence, Chandrayaan-2 CLASS, solar X-ray spectrum, GOES XRS, lunar surface composition

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1. Introduction

The major-element chemistry of the lunar surface records how the crust formed and evolved. Highland plagioclase is rich in Al and Ca; mare basalts are rich in Fe, Mg and Ti. Global maps of these elements constrain magma-ocean crystallisation, mare volcanism and impact mixing (Heiken et al., 1991). Orbital gamma-ray spectroscopy has provided global maps of Fe, Ti, Th and K, and model-dependent estimates of Mg, Al, Si and Ca, at a resolution of hundreds of kilometres (Lawrence et al., 1998; Prettyman et al., 2006).

X-ray fluorescence (XRF) excited by solar X-rays offers better spatial resolution and direct sensitivity to Mg, Al and Si (Jenkins, 1999; Beckhoff et al., 2006). The Chandrayaan-1 X-ray Spectrometer (C1XS) demonstrated the technique during a period of very low solar activity, when only a few flares gave usable spectra (Grande et al., 2009; Narendranath et al., 2011). Its successor on Chandrayaan-2, the Large Area Soft X-ray Spectrometer (CLASS), has a geometric area of 64 cm^2 and a footprint of a few tens of kilometres (Narendranath et al., 2014, 2020; Pillai et al., 2021). It has operated through the rise of solar cycle 25, and its Level-1 spectra are public.

Turning XRF line intensities into composition requires the spectrum of the incident solar X-rays. The fluorescence yield of each element depends on the solar flux above its absorption edge. A softer or harder solar spectrum therefore changes line ratios such as Mg/Si or Ca/Si even when the surface composition is unchanged. The Chandrayaan-2 Solar X-ray Monitor (XSM) measures this spectrum (Mithun et al., 2020), and the instrument team's analyses combine CLASS spectra with simultaneous XSM data. For users without an XSM-based workflow, the usual fallback is to assume a reference solar spectrum or to rely on line ratios. Ratios do not remove the dependence on the solar spectrum, because each element samples a different part of it.

In this paper we develop and validate a CLASS processing chain that can be reproduced entirely from public data, without XSM spectra. We take the solar excitation from the GOES X-Ray Sensor: its two-channel temperature, applied to isothermal CHIANTI spectra, gives a time-resolved solar spectrum for every CLASS exposure. Our specific contributions are:

1. a simplified fundamental-parameter forward model (Sherman primary fluorescence with matrix attenuation and incidence geometry, scattered solar spectrum, binned Gaussian detector response), fitted to

raw counts by Poisson maximum likelihood with per-spectrum uncertainties, goodness of fit and quality flags;

2. the identification of a solar-driven instrumental Al Kα component in CLASS spectra, present only when the spacecraft is sunlit, and a two-template background model that separates it from the particle background;
3. a quantified error budget from synthetic recovery tests, including the bias caused by errors in the solar temperature, which shows which element ratios can and cannot be recovered without a measured solar spectrum;
4. global Mg/Si and Al/Si maps from 24 monthly archives, with observation-count and uncertainty layers, validated against returned-sample compositions at nine landing sites and against Lunar Prospector GRS maps, using analysis choices fixed before validation.

We report element ratios relative to Si, not absolute abundances, and we state explicitly which quantities the data cannot constrain.

2. Data

2.1. CLASS Level-1 spectra

The Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) carries sixteen e2v CCD-236 swept-charge devices (SCDs) of 4 cm^2 each, behind a collimator with a field of view of about 14° full width (Narendranath et al., 2014; Pillai et al., 2021). We use the publicly released Level-1 product from the PRADAN portal of the Indian Space Science Data Centre. Each Level-1 file is an OGIP pulse-height table of 2048 channels holding the counts summed over the SCDs listed in the `SCD_USED` keyword for one 8 s integration, together with the four footprint corner coordinates, the solar incidence angle `SOLARANG`, the emission angle `EMISNANG`, the exposure time, detector temperature and the processing-software version.

We downloaded every monthly archive available to us: 24 archives (December 2020; January and February 2021; March 2022; March 2024; and every month from November 2024 to May 2026), totalling 1,515,239 Level-1 exposures. The archives are non-contiguous because CLASS observes only when the spacecraft attitude, power and data-volume budgets allow, and because releases on PRADAN lag the observations. The 2020–2022 months

Table 1: CLASS Level-1 header survey (4,800 files, 200 per month).

Keyword	Value / range
`DETCHANS`	2048 in all files
`GAIN`	13.5 eV/channel in all files (fitted in-flight value: Section 3.6)
`EXPOSURE`	8 s (99.4%); partial integrations 1.8 s to 7.8 s
`SCD_USED`	16 SCDs in 74%; 1–15 otherwise
`EMISNANG`	$< 3 \times 10^{-6}$° (nadir)
`SOLARANG`	0° to 166° (= `PHASEANG`)
`TEMP`	−54 °C to −22 °C (median −41 °C)
`SW_VERSN`	2.0 (Dec 2020–Feb 2021), 2.1 (later)
`RESPFILE`, `ANCRFILE`	empty

fall near the minimum of solar cycle 25 (Hathaway, 2015) and contribute few usable spectra (Section 7.1).

A survey of 4,800 headers sampled uniformly from all months (Table 1) shows that every file has 2048 channels and a nominal gain keyword of 13.5 eV/channel. No 1024-channel products occur in this release, so the 1024-channel branch of earlier processing is not used. The emission angle is $< 3 \times 10^{-6}$°, so the instrument views at nadir. Exposures are 8 s except for 0.6% partial integrations (1.8 s to 7.8 s). The number of SCDs summed varies: 26% of exposures use fewer than 16 devices, so the effective area changes from file to file. No response matrix (`RESPFILE`), ancillary response (`ANCRFILE`) or efficiency information ships with the release, and the background keyword is `NONE`.

2.2. GOES XRS solar soft X-ray data

CLASS has no on-board solar monitor of its own. We characterise the incident solar spectrum using the GOES X-Ray Sensor (XRS) 1-minute science-quality averages from NOAA/NCEI (GOES-16 for 2020–2022; GOES-18 for 2024–2026). The Moon and the Earth see the same solar disk to within a fraction of a degree. For each minute we derive the isothermal temperature T and emission measure from the ratio of the 0.05 nm to 0.4 nm and 0.1 nm to 0.8 nm channels (White et al., 2005), using the CHIANTI-based response tables with coronal abundances implemented in `sunkit-instruments` (The SunPy Community et al., 2020). Each CLASS exposure is assigned the values of the nearest GOES minute to its mid-time (tolerance 1 min).

Below the B1 flux level ($F_{0.1-0.8\,\mathrm{nm}} < 1 \times 10^{-7}\,\mathrm{W\,m^{-2}}$), the GOES-16 ratio temperature is unreliable: it is either pinned at the lower edge of the response table (2.8 MK) or diverges (Section 9). Exposures taken below B1 are therefore excluded from all maps. This threshold was fixed before any comparison with ground truth.

2.3. Calibration and validation data

For validation we use the best-estimate average soil compositions of the Apollo 11, 12, 14, 15, 16 and 17 and Luna 16, 20 and 24 landing sites, compiled from returned-sample analyses (Heiken et al., 1991, Table 7.15; Basu and Riegsecker, 1998) (Table 4), and the Lunar Prospector gamma-ray spectrometer (GRS) oxide maps of Prettyman et al. (2006). The latter are distributed by the PDS Geosciences Node on 5° equal-area pixels (Prettyman, 2012). Oxide weight fractions are converted to element weight ratios relative to Si using standard stoichiometric factors.

3. Forward model

We model the lunar regolith within one footprint as a flat, homogeneous, infinitely thick target. Its composition is parameterised by element-to-silicon weight ratios $r_i = C_i/C_{\mathrm{Si}}$, and the observed spectrum is the sum of four components: fluorescence lines, scattered solar continuum and lines, particle background, and a solar-driven instrumental component. All atomic parameters come from xraylib 4.2 (Schoonjans et al., 2011).

3.1. Composition closure

The fitted elements are Mg, Al, Ca, Ti and Fe, each expressed relative to Si. Each element is assumed to be present as its common lunar oxide (SiO_2, MgO, Al_2O_3, CaO, TiO_2, FeO). Minor oxides are fixed at typical soil values (Na_2O 0.5, K_2O 0.1, MnO 0.15, Cr_2O_3 0.2 wt%). Requiring the oxides to sum to unity fixes the absolute Si weight fraction:

$$C_{\mathrm{Si}} = \frac{1 - \sum_m w_m^{\mathrm{ox}}}{f_{\mathrm{Si}} + \sum_i r_i f_i}, \tag{1}$$

where f_i is the oxide-to-element mass factor and w_m^{ox} are the fixed minor-oxide fractions. Oxygen follows stoichiometrically, so it is not a free parameter. The matrix mass attenuation coefficient is $\mu(E) = \sum_j C_j\,\mu_j(E)$, with μ_j the total mass attenuation coefficient of element j.

3.2. Incident solar spectrum

The incident photon spectrum $S(E;T)$ is the isothermal CHIANTI spectrum, lines plus continuum with coronal abundances (Dere et al., 1997), evaluated with `sunkit-spex` at the GOES temperature of each exposure (Section 2.2). It is tabulated on a 5 eV grid from 1.01 keV to 20 keV at 61 logarithmically spaced temperatures between 2 and 40 MK, and interpolated in $\log T$. Its absolute normalisation is absorbed by a free amplitude.

Line emission dominates the excitation of the light elements. Between 3 and 15 MK, 63–80% of the photon flux above the Mg K edge, and 55–74% above the Si K edge, is in lines, mainly the He-like Mg XI, Si XIII and S XV complexes. A bremsstrahlung-only spectrum, as is often assumed when no solar monitor is available, therefore misrepresents the relative excitation of Mg, Al and Si (Section 9).

3.3. Primary fluorescence

For line ℓ (Kα or Kβ) of element i, the primary fluorescence intensity follows Sherman (1955):

$$P_{i\ell} = C_i\,\omega_{i,K}\,p_{i\ell}\int_{E_{K,i}}^{E_{\max}} \frac{S(E;T)\,\tau_{i,K}(E)}{\mu(E)/\sin\psi_1 + \mu(E_{i\ell})/\sin\psi_2}\,\mathrm{d}E, \tag{2}$$

where $\tau_{i,K}$ is the K-shell photoionisation cross-section, $\omega_{i,K}$ is the K-shell fluorescence yield, $p_{i\ell}$ is the radiative rate of the line, and $E_{K,i}$ is the K-edge energy. The incidence elevation is $\psi_1 = 90^\circ - $ `SOLARANG`, and the emergence elevation is $\psi_2 = 90^\circ$ (nadir). The product $\omega_{i,K}p_{i\ell}\tau_{i,K}$ reproduces the xraylib fluorescence-line cross-section to within 5%. The integral is evaluated on the 5 eV grid up to $E_{\max} = 15$ keV; a unit test shows this agrees with adaptive quadrature to within 1% and with a 1 eV grid to within 0.3%. Because the integrand depends on the composition through μ, Eq. (2) couples all elements. This coupling is the matrix effect; ratios of line intensities are therefore not equal to ratios of concentrations.

3.4. Secondary fluorescence

Fluorescence of element i excited by line m of element j ($E_{jm} > E_{K,i}$) is computed with the thick-target expression of de Boer (1990), which is linear in both C_i and C_j. Over 3 MK to 15 MK and both mare and highland compositions, the secondary contribution to every Kα line is at most 1.3% of the primary. We therefore omit it from the fitted model. We do, however,

include it when generating the synthetic spectra of Section 5, so the bias from omitting it is part of the measured recovery bias.

3.5. Scattered solar spectrum

Coherent and incoherent scattering of the solar spectrum by the regolith contributes a continuum, and at the GOES temperatures encountered, scattered solar lines as well:

$$\Sigma(E) = S(E;T)\,\frac{\sigma_s(E)}{\mu(E)/\sin\psi_1 + \mu(E)/\sin\psi_2}, \tag{3}$$

where $\sigma_s = \sum_j C_j(\sigma_{\mathrm{coh},j} + \sigma_{\mathrm{inc},j})$. The angular dependence of the differential scattering cross-sections is not modelled. Instead, the scattered component carries its own free amplitude B.

3.6. Detector response and energy scale

The centre energy of channel k is $E_k = g\,k + o$. Each line is distributed over channels by a Gaussian integrated over the channel boundaries $E_k^{\pm} = E_k \pm g/2$:

$$R_{\ell k} = \frac{\Phi\!\left(\frac{E_k^{+}-E_\ell}{\sigma_\ell}\right) - \Phi\!\left(\frac{E_k^{-}-E_\ell}{\sigma_\ell}\right)}{\mathrm{erf}(5/\sqrt{2})}, \qquad \sigma_\ell^2 = \sigma_0^2 + F\,w\,E_\ell, \tag{4}$$

where Φ is the standard normal cumulative distribution, truncated at $\pm 5\sigma$, $F = 0.115$ is the Fano factor of silicon and $w = 3.68\,\mathrm{eV}$. The normalisation conserves line area to 10^{-9} (unit test). The scattered spectrum, Eq. (3), is folded through the same response on the 5 eV grid.

The gain g, offset o and noise term σ_0 are determined for each month. We stack all 16-SCD dayside exposures of the month, subtract the particle background, and fit the Mg, Al, Si and Ca K lines jointly. For example, for December 2024 (75,707 exposures) we find $g = 13.74\,\mathrm{eV/channel}$, $o = 2.2\,\mathrm{eV}$ and $\sigma_0 = 74\,\mathrm{eV}$, i.e. 185 eV FWHM at Al Kα. The fitted gain is about 1.6% larger than the header value of 13.5 eV/channel. Using the header gain shifts the Si Kα centroid by about two channels. The per-month values are listed in Table A.5.

Each SCD views the surface through two optical-blocking filters, one on either side of the collimator, each made of 0.2 µm polyimide coated with 0.2 µm aluminium (Narendranath et al., 2014; Pillai et al., 2021). We apply

their X-ray transmission at the photon energy, before folding. Because the Al K edge (1.56 keV) lies between the Al and Si lines, the transmission is 0.91 for Al Kα but 0.69 for Si Kα (Mg Kα: 0.86). Ignoring the filters would inflate Al/Si by 31% and Mg/Si by 24%. The quantum efficiency of the SCDs themselves (depletion depth, surface dead layer) is not published and is not modelled (Section 9.4).

3.7. Background: particles and a solar-driven instrumental component

Earlier analyses of these data subtracted the mean nightside spectrum (`SOLARANG` $\geq 90°$). We find that this mean contains two physically distinct components (Fig. 3).

The first is the particle background, a smooth continuum with weak features. It dominates exposures taken while the spacecraft is in the lunar umbra, i.e. when `SOLARANG` exceeds $90° + \arccos[R_\mathrm{M}/(R_\mathrm{M} + h)] \approx 109°$ at $h = 100$ km. The second is a strong Al Kα line, together with a soft continuum and a feature near 6.7 keV. This second component appears only while the spacecraft itself is sunlit. In December 2024 the median background-corrected Al Kα rate of nightside exposures is 31.8 counts s^{-1} when the spacecraft is sunlit ($n = 8{,}555$), against 0.16 counts s^{-1} in the umbra ($n = 2{,}254$). Its rate correlates with solar proxies (Spearman $\rho = 0.75$ with the 0.6 keV to 0.9 keV counts) but not with the 8 keV to 20 keV particle rate ($\rho = 0.06$), and it varies on time scales shorter than 30 min. We interpret it as solar X-ray fluorescence of aluminium in the instrument structure. On the dayside it is superposed on the lunar Al line with a comparable magnitude.

We therefore build two templates per month, in units of counts s^{-1} per 16 SCDs. The first is b_k^p, the mean umbra spectrum. The second is b_k^s, the mean sunlit-nightside spectrum minus b_k^p scaled to the same 8 keV to 20 keV rate. Months with fewer than 100 umbra (or sunlit) nightside exposures use the December 2024 template shape. The instrumental amplitude cannot be predicted from GOES with useful accuracy: the GOES-predicted Al-exciting flux explains it only to a factor of $\approx$2 (Spearman $\rho = 0.39$), with an angle-dependent scale.

The geometry of Fig. 3 shows that the instrumental component requires the Sun to be behind the spacecraft. It is negligible already at `SOLARANG` $\approx$ 90–93°, and it is absent on the dayside, where the Sun is in front of the nadir-pointing instrument. Our primary model therefore fixes its amplitude to $\gamma = 0$ for all dayside spectra, so the background is $\beta\, b_k^\mathrm{p}$. We also ran an

alternative model in which γ is a free, non-negative parameter. It is reported in Section 8.4, together with the reason the primary model was chosen.

4. Spectral fitting

4.1. Likelihood

Each exposure is fitted on its raw counts d_k without smoothing, denoising or background subtraction. The fitted channels are those with $E_k \in [1.15, 7.5]\,\mathrm{keV}$. The lower limit avoids the Na Kα line at 1.04 keV and a steep excess below 1.1 keV that is present in every stacked spectrum. One additional bin holds the summed counts in 8 keV to 20 keV, where no lunar fluorescence is expected. That bin fixes the particle-background level without depending on the background shape. The expected counts are

$$m_k = A\Big[\sum_{\ell} \frac{P_\ell}{P_{\mathrm{ref}}} R_{\ell k} + B\,\frac{(\mathcal{F}\Sigma)_k}{\Sigma_{\mathrm{ref}}}\Big] + \beta\, b^{\mathrm{p}}_k\, t\, \frac{n_{\mathrm{SCD}}}{16}, \tag{5}$$

where t is the exposure, n_{SCD} is the number of SCDs summed, $\mathcal{F}$ is the detector folding of Eq. (4), and P_{ref}, Σ_{ref} are fixed reference normalisations. The parameter vector is $\theta = (A, r_{\mathrm{Mg}}, r_{\mathrm{Al}}, r_{\mathrm{Ca}}, B, \beta)$. In the alternative model of Section 8.4, the term $\gamma\, b^{\mathrm{s}}_k\, t\, n_{\mathrm{SCD}}/16$ is added with $\gamma \geq 0$ free. All components are non-negative, the ratios are bounded to $[0, 5]$, and $\beta \leq 10$.

Si is the reference element, so only ratios and one amplitude are fitted. This removes the scale degeneracy of fitting absolute concentrations together with a free global scale. Ti/Si and Fe/Si are held at global lunar-soil means (0.04 and 0.37) and enter only through the matrix. At GOES temperatures of 3–8 MK the solar flux above the Ti and Fe K edges is too small for an 8 s spectrum to constrain them; leaving them free made 92% of test fits ill-conditioned. We minimise the Cash statistic (Cash, 1979)

$$C = 2\sum_{k} \big[m_k - d_k + d_k \ln(d_k/m_k)\big] \tag{6}$$

with a trust-region reflective least-squares solver applied to the signed deviance residuals, whose sum of squares equals C.

4.2. Uncertainties, goodness of fit and quality flags

Parameter covariances are taken from the inverse Poisson Fisher matrix $\mathcal{I} = J^{\top}\mathrm{diag}(1/m)J$, where $J = \partial m/\partial\theta$ is computed by finite differences

at the optimum. We also record the correlation matrix and the condition number of its core block $(A, r_{\mathrm{Mg}}, r_{\mathrm{Al}}, B, \beta)$. For goodness of fit we compare C over the main window with its expectation and variance for the fitted model, computed exactly for Poisson counts (Kaastra, 2017): $z = (C - \langle C \rangle)/\sqrt{\mathrm{Var}\, C}$.

A spectrum enters the maps (quality = 0) only if all of the following hold:

- the fit converged;
- there are at least 300 counts in the 1.15 keV to 7.5 keV window;
- $|z| \leq 3$;
- neither Mg/Si nor Al/Si is at a bound;
- the core condition number is below 10^4;
- the geometry is valid ($0.5^\circ < \psi_1 \leq 90^\circ$, altitude > 0);
- a reliable GOES temperature exists (flux $\geq$ B1).

These criteria, the 2° mapping grid and the validation radii were fixed and logged before any comparison with reference data. The count threshold is varied (300, 1000, 3000) as a sensitivity test. No flux-percentile selection is applied.

4.3. Products

Ca/Si is fitted because it shapes the model between the Si and Ca lines, but it is not reported. The synthetic tests of Section 5 show it is biased even at the correct temperature and changes by an order of magnitude under realistic temperature errors. The products are Mg/Si and Al/Si weight ratios with per-footprint uncertainties.

5. Synthetic recovery tests

Before applying the fitter to real data, we test it on synthetic CLASS-like spectra.

Generator. The synthetic spectra are generated with the full forward model, including secondary fluorescence, which the fitted model omits. The calibration is the December 2024 one, the background uses the real December 2024 particle template ($\beta = 1$), and each spectrum is an 8 s, 16-SCD exposure with Poisson noise. Compositions are drawn uniformly from lunar ranges (Mg/Si 0.10–0.35, Al/Si 0.20–0.80, Ca/Si 0.30–0.60), with incidence angles of 10–80°. The scattered-solar amplitude is set to match the continuum level of the December 2024 stack.

Results at the correct temperature. The signal level is expressed as the number of Si Kα counts. For a 5 MK solar spectrum (the median GOES temperature of 2024–2026), Table 2 lists the median relative bias and robust (MAD-based) relative scatter of the recovered ratios. It also lists the width of the pull distribution, $(\hat{r} - r)/\hat{\sigma}_r$. Mg/Si and Al/Si are recovered with $|\text{bias}| \leq 4\%$ for every signal level, and with pulls of width 0.84–1.25, so the Fisher uncertainties are realistic. Omitting secondary fluorescence does not bias them. In the alternative model, where the generator adds the instrumental template ($\gamma = 0.5$) and the fit leaves γ free, the recovery is equally unbiased, but the Al/Si scatter at 300 counts roughly doubles. Ca/Si is biased by up to +50%, with 50–150% scatter, even when the temperature is exact, because a 5 MK spectrum barely excites Ca.

Temperature error. The main systematic is the solar temperature. We generate spectra at one temperature and fit them assuming another, to mimic error in the GOES temperature. With a true spectrum 30% hotter than assumed, Mg/Si is biased by −15 to −13% and Al/Si by +3 to +5%. With a true spectrum 20% cooler, the biases are +28 to +33% and +5 to +7%. Ca/Si changes by a factor of ∼9 or collapses to zero. These numbers set the systematic uncertainty quoted for the maps (Section 9.4). They also show why Ca/Si, Ti/Si and Fe/Si cannot be reported without a measured solar spectrum.

6. Mapping

Gridding. Footprints are placed on a 2° latitude–longitude grid. Each footprint quadrilateral is sampled with a 5×5 bilinear grid of points between its four corners, and its weight is shared equally among the grid cells containing those points. The footprint corners recorded in the Level-1 headers span a median of 25 km across track by 37 km along track. This is the collimator

Table 2: Synthetic recovery at a 5 MK solar temperature (80 spectra per level): median relative bias / robust relative scatter (%), and pull width.

Si Kα counts	300	1,000	3,000	10,000
Mg/Si bias / scatter (%)	+1 / 21	−4 / 10	−1 / 7	−1 / 5
Mg/Si pull width	0.84	0.86	1.13	1.25
Al/Si bias / scatter (%)	+2 / 16	0 / 8	−1 / 4	0 / 3
Al/Si pull width	0.91	0.97	0.93	1.14
Ca/Si bias / scatter (%)	+65 / 245	+38 / 127	+52 / 73	+27 / 55
Ca/Si pull width	0.87	0.90	1.08	1.82
Quality-0 fraction	1.00	1.00	1.00	1.00

field of view, about 25 km at 100 km altitude, plus roughly 13 km of ground-track motion during the 8 s integration. One degree is 30.3 km at the lunar equator, so a footprint overlaps one to four 2° cells.

Cell values and quality layers. Each cell value is the inverse-variance-weighted mean of the footprint ratios. Three quality layers accompany each map:

- the number of footprints contributing to the cell;
- the standard error of the weighted mean;
- the weighted scatter of the footprint ratios.

Coverage. We report coverage separately for all dayside footprints and for the quality-0 mapping set. We give it both as a fraction of grid cells and as a fraction of lunar surface area, where cells are weighted by cos(latitude). No smoothing, interpolation or infill enters any statistic, map value or validation number in this paper.

Resolution. The effective spatial resolution of a single measurement is set by the footprint. Its full extent is about 25×37 km, and the collimator response has a narrower FWHM, quoted as 12.5 km by Pillai et al. (2021). The 2° cells (61 km at the equator) set the resolution of the maps. The grid spacing should not be read as a resolution finer than either of these.

Table 3: Exposure counts at each pipeline stage (all 24 months).

Stage	Exposures
Level-1 files	1,515,239
Dayside (`SOLARANG` $< 90°$), valid geometry; all fitted	1,053,989
with reliable GOES temperature ($\geq$ B1)	1,017,226
fit converged	1,037,846
Mapping set (quality = 0, $\geq$300 counts, $\geq$ B1)	578,530

7. Results

7.1. Pipeline statistics

Table 3 traces the exposures through the pipeline. All 1,053,989 dayside exposures with valid geometry were fitted; no flux-percentile pre-selection was applied. 578,530 spectra (54.9% of the dayside set) pass every quality criterion and form the mapping set. The fraction passing varies strongly with solar activity. It ranges from 2% in February 2021, near solar minimum, where most spectra fall below the count threshold or below GOES B1, to 66% in February 2025 (Table A.5). The GOES temperatures of the mapped exposures have a median of 5.2 MK (5th–95th percentile 3.6–8.0 MK).

7.2. Fit examples

Figure 2 shows fits to a bright, a median and a faint dayside spectrum. The model reproduces the Mg, Al and Si K lines, the Ca K line and the scattered solar continuum and lines. In the faint spectrum the particle background dominates above 2 keV, and the uncertainties on Mg/Si and Al/Si become correspondingly large. Bright flare spectra (e.g. the top panel, $T = 16.8$ MK) often fail the goodness-of-fit criterion. Residual structure below 2 keV points to limitations of the isothermal solar spectrum and of the line-shape model, so the mapping set is weighted towards non-flaring and moderate-flare conditions.

7.3. Global maps

Figure 1 shows the Mg/Si and Al/Si maps with their observation-count and standard-error layers. Over the mapping set, the median footprint ratios are Mg/Si = 0.22 (5th–95th percentile 0.07–0.43) and Al/Si = 0.92 (0.49–1.38). The Mg/Si range lies within the range of returned lunar soils. The

Al/Si median sits above most returned soils, which points to a residual efficiency or calibration offset between the Al and Si lines (Section 9.4). An offset of this kind scales all values but does not affect the spatial pattern or the correlations.

Quality-0 footprints reach every 2° cell (100.00% of cells, 100.0% of the area), and 99.7% of the surface is covered by cells with at least five footprints. Coverage is complete because the 24 archives include polar orbits at all local times. It is not produced by interpolation. The polar cells have the fewest footprints and the largest standard errors.

The Mg/Si map shows the expected first-order dichotomy: higher values over the nearside maria and the South Pole–Aitken basin, lower values over the farside feldspathic highlands. The Al/Si map shows the complementary pattern at higher contrast. Individual maria (Imbrium, Serenitatis, Crisium, Tranquillitatis, Fecunditatis, Nubium, Humorum) and the South Pole–Aitken basin are resolved as low-Al/Si units against the feldspathic highlands. Residual north–south striping follows individual orbits. It reflects orbit-to-orbit differences in the solar state that the GOES isothermal temperature does not fully capture (Section 9).

8. Validation

The following choices were fixed and logged before the validation of the primary model was run:

- the matching radii (1, 2 and 5°, with 2° primary);
- the statistics: Pearson and Spearman coefficients with bootstrap 95% confidence intervals, leave-one-site-out ranges, and a 20° spatial block bootstrap for GRS;
- the criterion for claiming an element as validated: Pearson $r > 0$ with a 95% confidence interval excluding zero in both the landing-site and GRS tests.

The history of the primary model, including an earlier model whose validation motivated it, is given in Section 8.4.

Table 4: Landing-site validation at 2° radius: returned-soil weight ratios and CLASS weighted means ± standard error.

Site	Lat (°)	Lon (°E)	Mg/Si soil	Mg/Si CLASS	Al/Si soil	Al/Si CLASS
Apollo 11	0.67	23.47	0.238	0.311±0.005	0.365	0.679±0.008
Apollo 12	-3.01	-23.42	0.259	0.282±0.005	0.316	0.545±0.007
Apollo 14	-3.65	-17.47	0.252	0.324±0.003	0.410	0.639±0.005
Apollo 15	26.13	3.63	0.317	0.322±0.007	0.353	0.643±0.010
Apollo 16	8.97	15.50	0.163	0.292±0.006	0.687	0.758±0.010
Apollo 17	20.19	30.77	0.311	0.302±0.006	0.448	0.672±0.009
Luna 16	-0.68	56.30	0.272	0.314±0.004	0.415	0.662±0.006
Luna 20	3.53	56.55	0.280	0.300±0.004	0.560	0.811±0.008
Luna 24	12.71	62.20	0.276	0.308±0.004	0.322	0.649±0.006

8.1. Landing sites

Table 4 lists the returned-soil ratios (average soil compositions; Heiken et al., 1991, Table 7.15; Basu and Riegsecker, 1998) and the CLASS weighted means within 2° of each site.

- *Al/Si, 2°:* $r = 0.81$ (95% CI [0.48, 0.97]), Spearman $\rho = 0.78$ ($p = 0.01$), leave-one-out range 0.81–0.86.
- *Al/Si, other radii:* $r = 0.71$ (CI [0.51, 0.99]) at 1° and $r = 0.47$ at 5°.
- *Mg/Si, 2°:* $r = 0.38$ ([−0.50, 0.83]); $r = -0.64$ at 1° and 0.09 at 5°.

The CLASS site means have small statistical errors (Table 4). Their scatter about the soil values is therefore systematic. Footprints of 25×37 km average over terrain much larger than the sampled landing sites, and several sites lie on mare–highland contacts (Apollo 15, 16, 17; Luna 20). Al/Si passes the site criterion; Mg/Si does not.

8.2. Lunar Prospector GRS

All 1,790 5° equal-area GRS pixels contain at least 20 CLASS footprints. On these pixels:

- *Al/Si:* $r = 0.72$ (block-bootstrap 95% CI [0.66, 0.76]; Spearman $\rho = 0.60$);

- *Mg/Si:* $r = 0.61$ ([0.56, 0.66]; $\rho = 0.63$).

Both pass the GRS criterion (Fig. 4). The correlations are stable under the count-threshold test: Al/Si r = 0.72, 0.72 and 0.69, and Mg/Si r = 0.61, 0.58 and 0.58, for thresholds of 300, 1000 and 3000 counts (578,530, 143,961 and 11,259 spectra). Restricting to GOES $\geq$ B3 leaves them unchanged.

8.3. How close are these to the attainable limit?

Agreement with GRS is bounded by noise in both data sets. We split the mapping set into two halves by alternate months, map each half separately, and correlate the two maps. After Spearman–Brown correction, the CLASS pixel means have reliabilities of 0.97 (Al/Si) and 0.76 (Mg/Si). The GRS error matrices imply reliabilities of 0.78 and 0.73. The maximum attainable correlations are therefore $\sqrt{\rho_{\mathrm{CLASS}}\rho_{\mathrm{GRS}}} = 0.87$ for Al/Si and 0.75 for Mg/Si. The observed values reach about 80% of these ceilings. The GRS Mg and Al abundances are themselves model-dependent (Prettyman et al., 2006).

The landing-site test has little discriminating power at this scale. Applied to the same nine soil averages, using the 5° pixel containing each site, GRS itself gives $r = -0.34$ for Al/Si and -0.11 for Mg/Si. Soil averages sample metres to kilometres around each landing point, whereas orbital pixels average tens to hundreds of kilometres (Basu and Riegsecker, 1998). The failure of Mg/Si in this test therefore says little about the map, while the success of Al/Si is notable.

Comparison with the earlier processing. The earlier processing of the same data (centre-point binning of denoised least-squares fits without a solar model) gives $r = 0.19$ (Al/Si) and 0.48 (Mg/Si) on the same GRS pixels.

Weighted median. Replacing the weighted mean by a weighted median changes little: GRS $r = 0.71$ (Al/Si) and 0.58 (Mg/Si). Figure 5 shows the weighted-median maps.

8.4. Alternative model and the history of the primary model

The analysis was first run with the instrumental amplitude γ free (alternative model, 429,217 mapped spectra). Under the same pre-declared rules, that model gave the following.

- *GRS:* Al/Si $r = 0.35$ ([0.26, 0.44]) and Mg/Si $r = 0.60$ ([0.54, 0.65]).
- *Landing sites (2°):* Al/Si $r = 0.43$ and Mg/Si 0.37. Neither element was claimed under the criterion.

- *Diagnosis:* the Al/Si reliability was only 0.42 (ceiling 0.57). In a fixed farside region, the monthly median Al/Si anticorrelated with the fraction of spectra having a non-zero fitted γ (Spearman -0.88): in some months the free γ absorbed part of the lunar Al line.

- *Post-hoc check:* restricting that model to spectra with $\gamma < 0.05$ (51%) gave GRS Al/Si $r = 0.72$.

Because the nightside geometry (Fig. 3) independently indicates that the component is absent on the dayside, we then specified the primary model ($\gamma \equiv 0$ on the dayside). We logged it, together with the commitment to report its outcome whatever it was, *before* running it on the full data set. All other choices are identical to the alternative model.

During development we also corrected one defect after a first validation. The boundary threshold used to flag pinned ratios was raised from 10^{-6} to 10^{-3}, because 254 fits pinned near zero had collapsed Fisher uncertainties and dominated the inverse-variance weights. Before that fix, the alternative model gave GRS Al/Si $r = 0.21$.

The primary model was thus motivated by the diagnosis of the alternative model's validation, and readers should weigh it accordingly. Its specification was nonetheless fixed in advance, and it rests on a physical argument that does not involve the reference data.

8.5. Outcome

Under the pre-registered rule, Al/Si is claimed as validated: it passes both the GRS test ($r = 0.72$) and the landing-site test ($r = 0.81$). Mg/Si passes the GRS test ($r = 0.61$) but not the nine-site test, which GRS itself does not pass. We therefore present Al/Si as a validated relative map. We present Mg/Si as a relative map whose large-scale pattern agrees with GRS, without a claim of site-level accuracy.

9. Discussion

9.1. Why the solar spectrum matters for line ratios

Ratios to Si are often treated as insensitive to the incident spectrum. Our forward model shows that they are not. Each K line is excited by the part of the solar spectrum above its own edge (Mg 1.31 keV, Si 1.84 keV, Ca 4.04 keV), and at coronal temperatures that part is dominated by a few He-like line complexes whose relative strengths change steeply with temperature.

In our synthetic tests, a 30% hotter spectrum than assumed biases Mg/Si by −15 to −13% and changes Ca/Si by an order of magnitude. Refitting real data with the GOES temperature scaled by 0.8 and 1.2 changes the median Mg/Si by −25% and +27%, and Al/Si by −9% and +8%.

Before the solar lines were included, a continuum-only reference spectrum with any single temperature produced Mg/Si values of 0.3–0.8 together with Ca/Si of 0.004–0.5, never simultaneously lunar. The implausible ratios of earlier CLASS processing (Mg/Si ≈ 1.4, Ca/Si ≈ 0.01) have the same origin. A time-resolved description of the solar spectrum is therefore not an optional refinement but a requirement, even for ratio maps.

9.2. GOES as a proxy for the solar spectrum

GOES provides the solar state for every CLASS exposure from public data. Its limitations are well known. The two-channel temperature is a single isothermal value. It is weighted towards the hottest plasma. Below the B1 level it is unreliable, and we found it pinned to the edge of the response table or diverging there. The GOES-16 and GOES-18 temperatures agree to 0.5% in their overlap.

The real corona is multi-thermal. The isothermal approximation underpredicts the flux above 4 keV at the temperatures typical of our data. This is why Ca/Si is biased even in self-consistent synthetic tests, and why Ti and Fe cannot be constrained at all. The residual orbit-to-orbit striping in the maps is consistent with solar-state errors of this kind. XSM spectra measured simultaneously with CLASS (Mithun et al., 2020) would remove this limitation, and our model accepts any tabulated $S(E)$, so the extension is direct.

9.3. The instrumental aluminium component

The Al Kα excess in sunlit nightside spectra has a sharp onset near `SOLARANG` ≈ 93° and disappears at the shadow boundary (Fig. 3). It must therefore be excited by solar X-rays that reach aluminium in the instrument (filters, frames, housing; Pillai et al., 2021) when the Sun is behind the spacecraft. On the dayside the Sun is in front of the nadir-pointing instrument, and the component is absent.

Mishandling it is costly. Subtracting a month-averaged nightside spectrum, as earlier processing did, removes a time-variable amount of Al. Fitting its amplitude freely, as in our alternative model, lets it absorb lunar Al in some months; Al/Si reliability then drops from 0.97 to 0.42 and the

GRS correlation from 0.72 to 0.35. Other XRF instruments with aluminium structures or filters may show the same effect, and it can be identified with the same nightside test.

9.4. Limitations

1. *No measured solar spectrum.* The solar spectrum is isothermal at the GOES temperature. In synthetic tests, temperature errors of +30% and −20% bias Mg/Si by −15 to −13% and +28 to +33%, and Al/Si by +3 to +5% and +5 to +7%. Scaling the GOES temperature of real data by 0.8 and 1.2 changes Mg/Si by −25% and +27% (Section 5). Ca/Si, Ti/Si and Fe/Si are not retrievable.
2. *Relative, not absolute.* The products are weight ratios to Si. No absolute calibration is attempted, and the amplitude A absorbs the solar flux, area and geometry.
3. *Detector efficiency.* The published filter transmission is applied. The SCD quantum efficiency and any surface dead layer are not modelled, because no curve is published. This mainly affects the Si line relative to Mg and Al, and probably explains why the median Al/Si (0.92) is higher than most returned soils. Such an offset rescales the maps but leaves their pattern and correlations unchanged.
4. *Physical effects not modelled.* Particle-size and roughness effects, phase-angle dependence beyond the plane-parallel geometry, and the angular dependence of scattering are not modelled.
5. *Spatial resolution.* Footprints are 25×37 km. The maps are 2° and should not be interpreted at finer scales.
6. *Small validation sample.* Nine landing sites, sampled at the scale of a footprint, give a test with little statistical power. The GRS comparison is limited by the GRS resolution and by its own model dependence for Mg and Al.

10. Conclusions

We have built a CLASS processing chain from public data only. It combines a simplified fundamental-parameter model, a CHIANTI solar spectrum at the GOES temperature of each exposure, the published filter transmission, a two-template background model and a Poisson likelihood fit with per-spectrum uncertainties and quality flags. Applied to 1,515,239 Level-1

exposures from 24 monthly archives, it gives 578,530 quality-controlled spectra and global 2° maps of Mg/Si and Al/Si with complete observed coverage. The main findings are:

1. Element/Si ratios depend strongly on the incident solar spectrum. Solar line emission carries most of the flux that excites Mg, Al and Si, and an isothermal spectrum at the GOES temperature brings the retrieved Mg/Si into the lunar range.
2. CLASS spectra contain a solar-driven instrumental Al Kα component when the Sun illuminates the spacecraft from behind. It must be kept out of the dayside background model, because either subtracting it as part of a mean nightside spectrum or fitting it freely corrupts Al/Si.
3. Without a measured solar spectrum, only Mg/Si and Al/Si can be recovered. Ca/Si, Ti/Si and Fe/Si are dominated by solar-spectrum systematics.
4. With the instrumental component excluded from the dayside model, Al/Si resolves individual maria and is validated against both Lunar Prospector GRS ($r = 0.72$, about 80% of the noise-limited ceiling) and nine landing-site soils ($r = 0.81$). Mg/Si agrees with GRS ($r = 0.61$); its landing-site test is inconclusive, as it is for GRS itself. The earlier processing of the same data reached $r = 0.19$ and 0.48 against GRS.

The code, per-spectrum results and maps are public. The most valuable next step is to replace the GOES-isothermal spectrum with simultaneous Chandrayaan-2 XSM spectra, which the model accepts without modification.

Appendix A. Per-month statistics and calibration

Data availability

CLASS Level-1 data are available from the ISRO PRADAN portal (https://pradan.issdc.gov.in/ch2/). GOES XRS 1-minute science-quality data are available from NOAA/NCEI (https://www.ncei.noaa.gov/products/goes-r-extreme-ultraviolet-xray-irradiance). The Lunar Prospector GRS elemental abundances are available from the PDS Geosciences Node (data set LP-L-GRS-5-ELEM-ABUNDANCE-V1.0). The processing code, the per-footprint results of the mapping set, the maps and the scripts that generate every number and figure in this paper are available at https:

Table A.5: Per-month exposure counts, mapping-set size, fitted energy-scale parameters and particle-template source.

Month	Files	Dayside	Mapped	g (eV/ch)	o (eV)	$\mathrm{FWHM}_{1.49}$ (eV)	Umbra template
2020-12	3,931	3,051	94	13.379	29.3	107	own
2021-01	4,634	3,501	0	13.383	13.7	108	own
2021-02	40,247	27,555	546	13.896	-24.4	92	own
2022-03	5,896	2,793	731	13.717	1.1	119	Dec 2024
2024-03	15,076	7,939	1,982	13.678	8.6	119	Dec 2024
2024-11	104,125	74,962	51,815	13.759	1.1	165	own
2024-12	97,345	75,720	42,704	13.737	2.2	185	own
2025-01	51,298	38,780	27,580	13.740	3.1	162	own
2025-02	92,570	63,780	44,826	13.739	3.6	156	own
2025-03	40,571	20,519	4,606	13.765	0.7	145	own
2025-04	81,585	48,469	18,672	13.839	-6.5	145	own
2025-05	94,521	65,910	39,426	13.634	16.1	156	own
2025-06	97,669	68,095	48,494	13.819	-5.8	168	own
2025-07	90,614	67,190	44,079	13.641	15.7	161	own
2025-08	97,593	66,706	39,325	13.821	-5.9	156	own
2025-09	40,747	23,210	11,039	13.898	-13.2	151	own
2025-10	27,237	16,137	2,018	13.684	9.4	169	own
2025-11	105,203	74,667	41,296	13.700	8.0	169	own
2025-12	92,318	71,126	44,952	13.747	3.3	176	own
2026-01	85,725	65,896	43,891	13.856	-8.1	176	own
2026-02	84,922	61,436	24,301	13.738	7.0	156	own
2026-03	31,184	18,295	2,913	13.669	15.1	160	own
2026-04	29,064	16,453	5,528	13.821	0.8	148	own
2026-05	101,164	71,799	37,712	13.766	7.9	155	own

//github.com/GeeteshPaidi/SomaDhatuNirupana and archived on Zenodo (https://doi.org/10.5281/zenodo.21993207).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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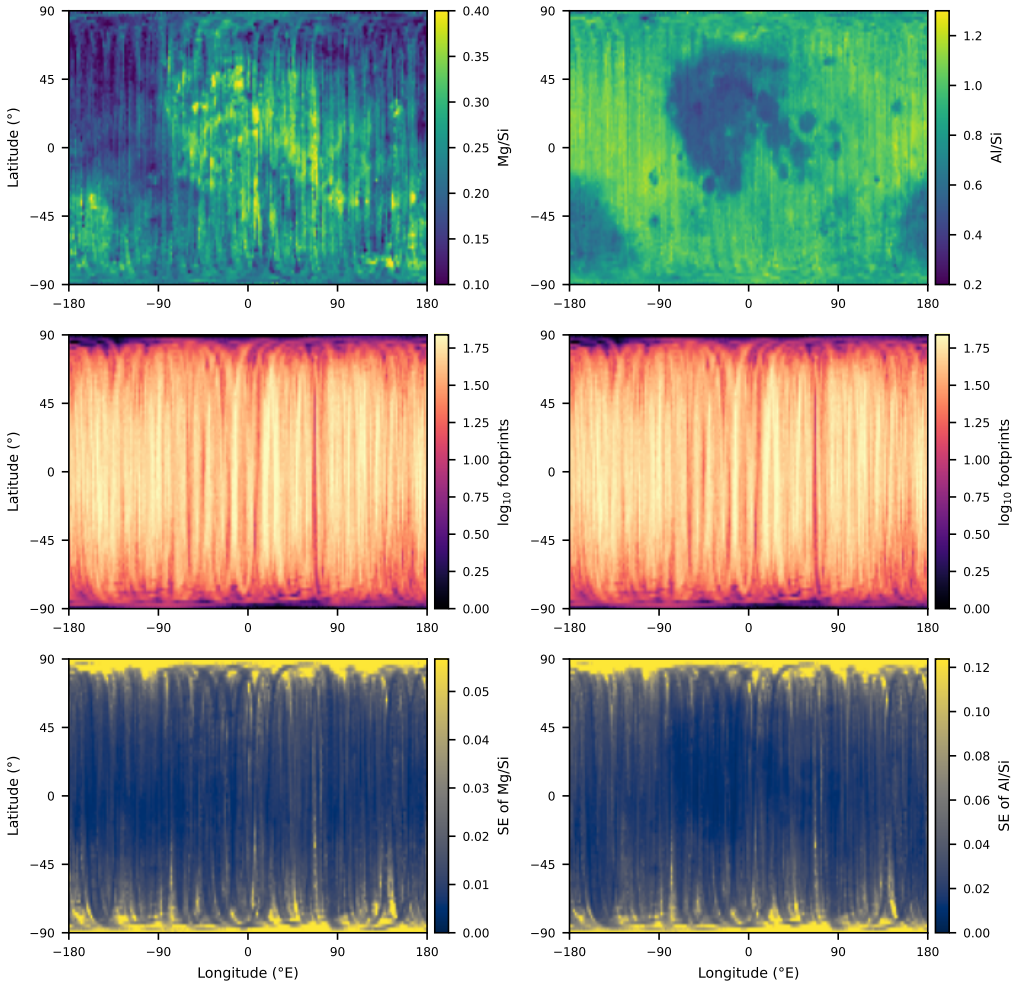


Figure 1: Mg/Si (left) and Al/Si (right) weight ratios on a 2° grid (inverse-variance-weighted means of quality-0 footprints; no smoothing or infill), with the number of footprints per cell (middle) and the standard error of the cell mean (bottom).

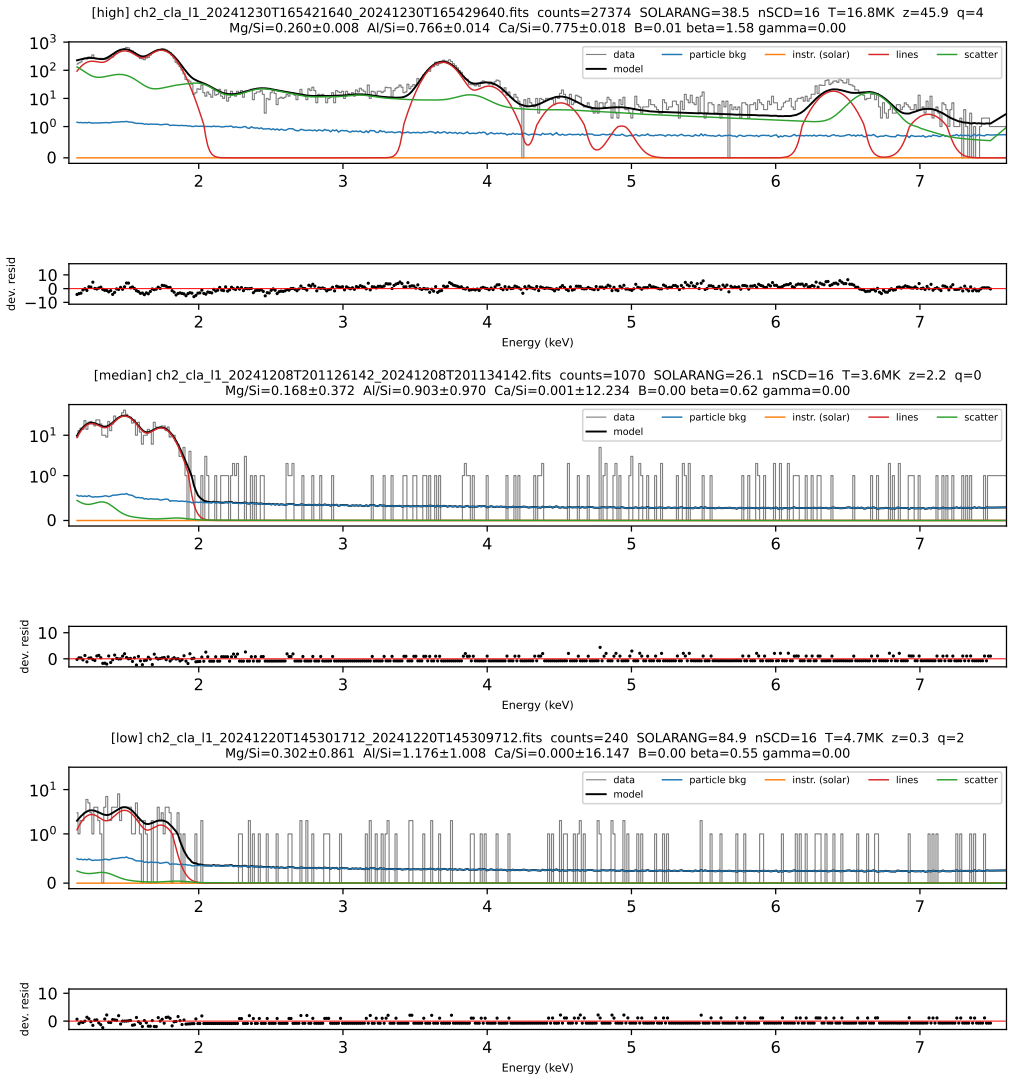


Figure 2: Example fits (December 2024) to a bright, a median and a faint dayside spectrum: data (grey), total model (black), fluorescence lines (red), scattered solar spectrum (green), and particle background (blue), with deviance residuals.

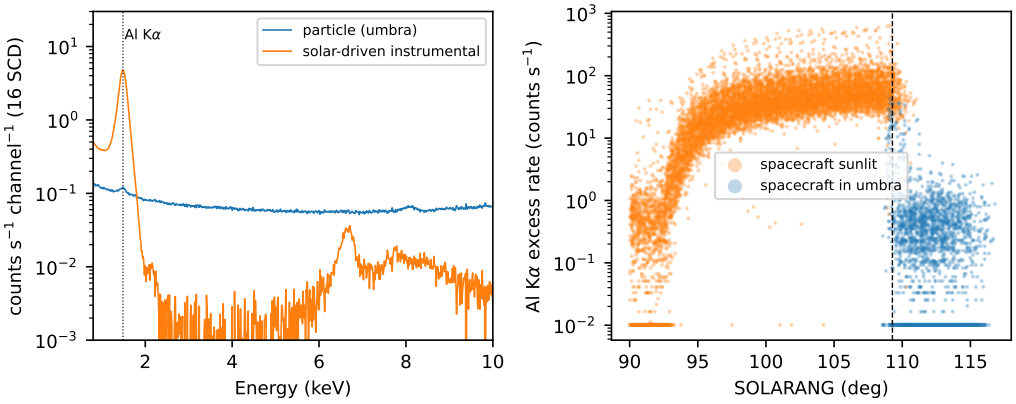


Figure 3: Left: December 2024 particle-background template (spacecraft in the lunar umbra) and solar-driven instrumental template (sunlit nightside minus particle background). Right: Al Kα excess rate of nightside exposures against `SOLARANG`; the dashed line is the shadow boundary of the spacecraft.

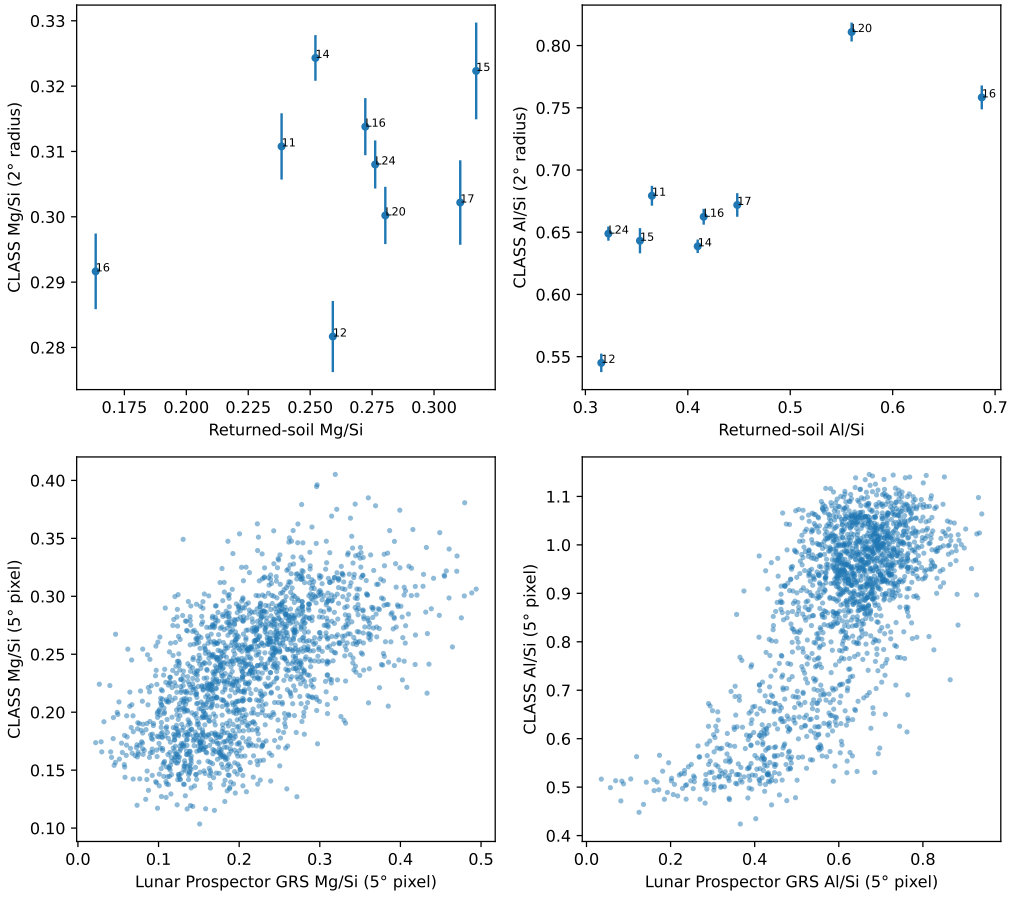


Figure 4: Top: CLASS ratios within 2° of the landing sites against returned-soil ratios. Bottom: CLASS against Lunar Prospector GRS ratios on 5° equal-area pixels.

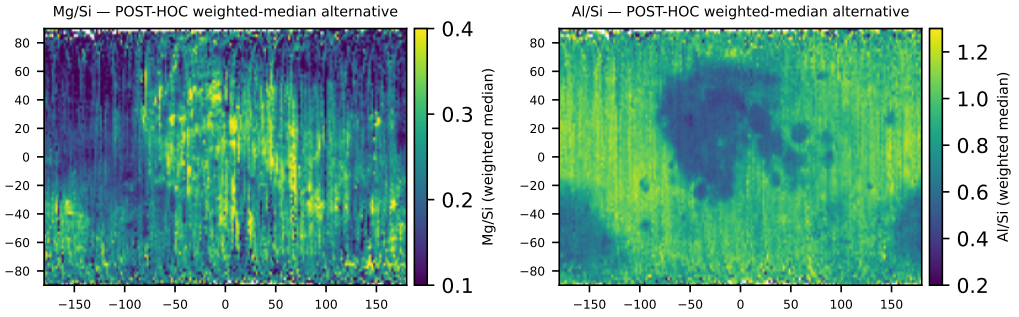


Figure 5: Alternative cell statistic (not the primary product): inverse-variance-weighted median of footprint ratios per 2° cell (footprint centres).